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Analytic Neural Network Gaussian Process Enabled Chance-Constrained Voltage Regulation for Active Distribution Systems with PVs, Batteries and EVs

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Abstract—This paper proposes an analytic neural network Gaussian process (NNGP)-based chance-constrained real-time voltage regulation method for active distribution systems with photovoltaics (PVs), batteries, and electric vehicles (EVs). NNGP can utilize historical measurement data to achieve real-time probabilistic node voltage estimation through Bayesian inference. Then, NNGP is fully analytically embedded into the optimal power flow model to perform voltage regulation and adapt to various topological changes. The uncertainties of voltage estimations are easily considered via the chance constraint, and it has been shown that the adoption of this chance constraint can significantly improve the reliability of voltage regulation under various scenarios. The comparison results with other methods, carried out on a real 759-node distribution system located in western Colorado, U.S., show that the proposed method can achieve accurate voltage estimation across different topologies and reliably perform voltage regulation considering PVs, batteries, and EVs.

Index Terms—Neural network Gaussian process, probabilistic voltage estimation, Bayesian inference, chance constraint, voltage regulation, PVs, EVs, active distribution systems.

NOMENCLATURE

Abbreviations

ADS	Active Distribution System
CRPS	Continuous Ranked Probability Score
DBN	Deep Belief Network
DER	Distributed Energy Resource
DGP	Deep Gaussian Process
DNN	Deep Neural Network
EV	Electric Vehicle
FA	False Alarm rate
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GP	Gaussian Process
GPU	Graphics Processing Unit
i.i.d.	independently and identically distributed
MAE	Mean Absolute Error
MD	Miss Detection rate
MPC	Model Predictive Control

MPPT	Maximum Power Point Tracking
NNGP	Neural Network Gaussian Process
OLTC	On-Load Tap Changer
OPF	Optimal Power Flow
PDIPM	Primal-Dual Interior-Point Method
PV	Photovoltaic
RL	Reinforcement Learning
SC	Switchable Capacitor
SVC	Static VAR Compensator
V2G	Vehicle-to-Grid
NVO	Number of Voltage Outside the κ

Notations

$\gamma, \gamma_{PV}/\gamma_V$	Coefficient to set different confidence probabilities, weight of the PV/voltage optimization term
ε	Preset confidence probability
ζ	Iteration count of PDIPM
$\theta_{x,x'}$	Angle between the inputs x and x'
κ	Voltage safety range $\kappa = [0.95, 1.05]$
Λ	Decision variables of the OPF or voltage regulation strategy
$\bar{\mu}/\bar{K}/\bar{\sigma}$	Mean/covariance/standard deviation of $\bar{v}$
σ_w^2/σ_b^2	Variances of the weight and bias parameters
ϕ	Pointwise nonlinearities function, i.e., activation function, and is set to ReLU
$A/B/R/T$	Nodes/battery/PV/test set
$\mathcal{C}$	Set of critical nodes that require strict voltage constraints
$\mathcal{D}$	Historical data $\mathcal{D} = \{(x^1, v^1), \dots, (x^n, v^n)\}$
d_{in}/d_{out}	Dimension of network input/output
$\mathbb{E}$	Mean function
e	Battery stored energy
F_ϕ	Recursive function between K^l and K^{l-1} , whose form depends only on the ϕ
$F(\bar{v}_{ij})$	Cumulative distribution function of $\bar{v}_{ij}$ with mean $\bar{\mu}$ and covariance $\bar{K}$
g_j/b_j	Conductance/susceptance at node j
$H(\bar{v}_{ij} - v_{ij}^*)$	Heaviside function and takes the value 0 when $\bar{v}_{ij} < v_{ij}^*$, and the value 1 otherwise
$\mathbb{I}_n$	$n \times n$ identity matrix
J/H	Jacobian/Hessian matrix
$J(\theta)$	Angular dependence

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i_{ij}	Current of line ij
$\mathbf{K}$	Covariance matrix
k	Topology of the system
$\mathbb{M}$	Data-driven estimation methods
$\max/\min$	Superscript for the upper and lower limits of $p/q/e/v$
$\mathcal{N}$	Normal distribution function
$N_n/N_s/N_t$	Number of nodes/samples/samples in the $\mathcal{T}$
N_l	Width of the l -th layer of network
N_v	Number of nodes that are at risk of voltage violations (the dimension of $\mathcal{V}$)
$\mathbb{P}$	Probability function
$\mathcal{P}/\mathcal{S}$	Set of nodes whose parent/child node are j
$\mathcal{P}$	Parameters of $\mathbb{M}$
p	Active power generation or demand of PV/Battery/EV/Load
$p^{\text{PV,MPPT}}$	Maximum value of p^{PV} according to MPPT
p_{jk}/q_{jk}	Active/reactive power flow of the line jk
q	Reactive power generation or demand of PV/Load/Other
r_{ij}/x_{ij}	Resistance/reactance of line ij
s^{PV}	Capacity of PV
$t, t_0, \Delta t$	Time instant, start time of the measured data, time interval between two voltage regulations (usually 15min or 1h)
$\mathcal{V}$	Set of nodes that have recorded voltage violations ($\bar{v} \notin \kappa$) according to historical data
$v, \bar{v}, v^*$	Voltage, estimated node voltage vector by data-driven methods, true labels of voltage
W_{ij}^l/b_i^l	Weight and bias parameters
$x_j^l(x)/z_i^l(x)$	Post-nonlinearity/post-affine transformation vector
x^0/z^L	Input/output of the network ($x^0 \in \mathbb{R}^{d_{\text{in}}}/z^L \in \mathbb{R}^{d_{\text{out}}}$)

I. INTRODUCTION

THE widespread growth of DERs and the emergence of new loads have caused voltage fluctuations in the ADSs and introduced uncertainties, making it difficult to address these issues effectively via traditional methods [1]. This section will discuss the necessity of the study from three aspects: voltage estimation methods, devices involved in voltage regulation, and voltage regulation methods.

In voltage regulation, it is necessary to repeatedly estimate the voltage performance corresponding to different regulation strategies. Most existing voltage estimation methods rely on physical models of the distribution network, which assume that accurate physical parameters, primarily the line impedances, of the network can be obtained. However, this is difficult to meet in practice due to mismatches between calculated values based on manufacturer's cable specifications and actual values [2]. Thanks to the advanced data measurement and communication technology in ADSs, enormous amounts of historical operation data can be collected, which includes the true physical models of ADSs [3]. Moreover, ADSs have flexible topology adjustment capabilities, where the topology is either known or can be

accurately derived based on measured data [4], [5]. Therefore, by utilizing historical measurement data and topological information, various data-driven voltage estimation methods have been proposed, such as random forests [6], long short-term memory network [7], GCN [8], etc. Among these data-driven methods, GCN can directly handle graph-structured data to address topological changes, while other methods tackle this issue by adding the on or off states of switches in the input data or training different models for different topologies. Training GCNs requires substantial computational resources, and other methods may introduce additional errors. Additionally, all these methods cannot handle new, unseen topologies unless the models are retrained. Data-driven methods also inevitably introduce additional estimation errors due to model bias or measurement noise. To quantify the estimation errors of these methods, probabilistic voltage estimation methods are proposed, including dropout [9], probabilistic extreme learning machine [10], ensemble learning [11], GP [12], etc. However, dealing with both voltage estimation uncertainties and topological changes poses a challenge for existing methods. The NNGP can directly perform Bayesian inference based on input data, eliminating the need for the training process required by conventional neural networks [13]. Therefore, it can conduct voltage inference based on historical measurement data corresponding to real-time measured or derived topology and quantify voltage estimation uncertainties.

Voltage regulation involves many devices, but their activation times and frequencies differ. Traditionally, voltage regulation is accomplished through adjustments of OLTCs, SCs, and SVCs. However, the control strategies of these devices are generally determined in day-ahead scheduling based on the estimation of PV outputs and load demands. Due to their slow response and discrete control, these devices cannot provide accurate real-time voltage regulation [14]. By contrast, inverter-based PVs can provide a fast power response for voltage regulation [15]. EVs, which have significantly increased in recent years, are highly likely to be plugged in for charging at night, thereby drawing a substantial amount of electricity from the ADSs, causing local voltage violations [16]. The temporal mismatch between PVs and EVs may cause severe generation-load imbalances and voltage fluctuation [17]. With the development of energy storage, the battery is connected with PV to form a PV-battery system to address this issue [18]. For example, in [19], battery energy storage systems are used to mitigate voltage rise during the peak PV generation and to address voltage drop during the peak load periods. In some studies, to reduce the high cost associated with energy storage, increased EVs are used to substitute for energy storage through the V2G mode [20]. However, the implementation of V2G is fraught with various challenges, including the capacity of EVs participating in V2G, the uncertainty of EVs charging and discharging, the willingness of EV owners, and the lifespan of EV batteries. Therefore, instead of considering V2G, this paper focuses on deploying standalone battery systems.

Voltage regulation is essentially an OPF problem. Due to the computational burden and complexity associated with nonlinear non-convex OPF models, most existing research on voltage regulation in distribution networks is based on approx-

imate flow models, including linearized distribution system power flow [21], convex quadratic programming [22], second-order cone relaxation [23], etc. Although these methods are reasonable in most cases, they may lead to control deviations at local nodes under extreme conditions. Additionally, the rise of RL in recent years has led to the development of a series of RL-based voltage regulation methods [24]. For example, [25] proposes a multi-agent RL-based approach for voltage regulation in distribution systems, enabling co-ordinated control of PV inverters through local information. Furthermore, deep learning-based surrogate models have been integrated into RL frameworks for voltage estimation [8], [26]. This integration partially addresses the issue of discrepancies between physical models and the real-world model. However, RL-based methods also face challenges, such as difficulties in strictly ensuring that constraints are met, as well as the inability to handle new and unseen scenarios in the training data. To address control errors caused by various uncertainties, probabilistic voltage regulation methods have been proposed, such as probabilistic PV output scenarios-based constrained OPF [27], probability density function-based static voltage stability margin control [28], nonparametric Bayesian framework-based probabilistic OPF [29], etc. These methods are based on modeling uncertainty and often involve approximations, which may not accurately reflect the true distribution of uncertainties. This paper conducts real-time voltage regulation based on measurement data, eliminating the need to model sources of uncertainty. Therefore, voltage regulation errors arise solely from data measurement and voltage estimation. NNGP can provide accurate voltage estimations based on historical and real-time measurement data, and quantify the uncertainties caused by measurement and estimation. By fully integrating NNGP into the nonlinear non-convex OPF, accurate voltage regulation can be obtained.

This paper proposes an analytic NNGP-based chance-constrained real-time voltage regulation method. The main contributions are summarized as follows:

- 1) NNGP is tailored to achieve real-time voltage estimations and uncertainty quantification while being robust to topology changes. Since it does not require pre-training, NNGP can utilize specific historical data for different network topologies to directly perform Bayesian inference of voltage, effectively addressing the limitations of traditional neural networks in handling topology changes. Moreover, compared to other deep learning methods, NNGP can significantly enhance the accuracy of voltage estimations.
- 2) The derived NNGP is analytically embedded into the OPF to obtain voltage regulation strategies without accurate ADS model assumptions compared to existing OPF-based methods. This is achieved in two ways: (1) Add NNGP-based voltage estimation into the objective function to optimize the voltages of all nodes that have voltage violation risks towards reference value, and (2) Use NNGP-based probabilistic voltage estimation distributions to establish voltage chance constraints for critical user nodes, achieving uncertainty quantification

and reducing safety risks. In particular, by deriving the Jacobian and Hessian matrices of the NNGP output (voltage) concerning the input variables, the voltage regulation strategy can be analytically solved via the PDIPM. The proposed estimation and regulation methods can be implemented in real-time, allowing for flexible adjustment of the objective function, critical user nodes, confidence probabilities, and constraint ranges in each regulation interval.

- 3) The validations are carried out on a real 759-node distribution system in western Colorado, U.S., unlike most existing methods, which are typically tested on IEEE benchmark systems and may not accurately reflect practical scenarios.

The rest of the paper is organized as follows. Section II presents the objective and constraints of the traditional voltage regulation framework. Section III shows the proposed NNGP-based chance-constrained voltage regulation method. Results are presented and analyzed in Section IV, and finally, Section V concludes the paper.

II. PROBLEM STATEMENT

For distribution system voltage regulation, the OPF-based approach is widely used to find the optimal dispatch subject to various constraints, which is mathematically formulated as follows [30], [31] :

A. Objective Function

The objective function is to minimize the PV active power curtailment (PV term) and optimize the voltages of all nodes that have the voltage violation risks towards reference value 1.0 p.u. (voltage term) [32], [33].

$$\min \gamma_{PV} \sum_{r \in \mathcal{R}} (p_{r,t}^{PV,MPPT} - p_{r,t}^{PV}) + \gamma_V \sum_{j \in \mathcal{V}} (v_{j,t} - 1)^2 \quad (1)$$

B. Power Flow Equations

The power flow equations for the radial distribution network can be represented as [31], [34]

$$p_{j,t} = \sum_{k \in \mathcal{P}} p_{jk,t} - \sum_{i \in \mathcal{S}} (p_{ij,t} - r_{ij} i_{ij,t}^2) + g_j v_{j,t}^2, j \in \mathcal{A} \quad (2)$$

$$q_{j,t} = \sum_{k \in \mathcal{P}} q_{jk,t} - \sum_{i \in \mathcal{S}} (q_{ij,t} - x_{ij} i_{ij,t}^2) + b_j v_{j,t}^2, j \in \mathcal{A} \quad (3)$$

$$v_{j,t}^2 = v_{i,t}^2 - 2(p_{ij,t} r_{ij} + q_{ij,t} x_{ij}) + i_{ij,t}^2 (r_{ij}^2 + x_{ij}^2), j \in \mathcal{A} \quad (4)$$

$$p_{j,t} = p_{j,t}^{PV} + p_{j,t}^{Battery} - p_{j,t}^{EV} - p_{j,t}^{Load}, j \in \mathcal{A} \quad (5)$$

$$q_{j,t} = q_{j,t}^{PV} - q_{j,t}^{Load} + q_{j,t}^{Other}, j \in \mathcal{A} \quad (6)$$

where $p_{j,t}^{PV}$, $p_{j,t}^{Battery}$, $p_{j,t}^{EV}$ and $p_{j,t}^{Load}$ represent the active power related to PV generation, battery charging and discharging, EV charging, and load demand, respectively, at node j at time t ; $q_{j,t}^{PV}$, $q_{j,t}^{Load}$ and $q_{j,t}^{Other}$ represent the reactive power related to PV generation, load demand and other devices such as OLTCs, SCs and SVCs, respectively, at node j at time t . It's important to note that the control strategies for

OLTCs, SCs, and SVCs are generally determined during the day-ahead scheduling. Therefore, $q_{j,t}^{\text{Other}}$ is a fixed value in real-time voltage regulation. Additionally, if the battery is discharging, $p_{j,t}^{\text{Battery}}$ is positive, and if the battery is charging, $p_{j,t}^{\text{Battery}}$ is negative. Moreover, if node j lacks PV, battery, EV, or load components, the corresponding active and reactive power values will be 0.

C. PV Constraints

The PV constraints include active power, reactive power, and capacity limitations:

$$0 \leq p_{r,t}^{\text{PV}} \leq p_r^{\text{PV,MPPT}}, r \in \mathcal{R} \quad (7)$$

$$q_r^{\text{PV,min}} \leq q_{r,t}^{\text{PV}} \leq q_r^{\text{PV,max}}, r \in \mathcal{R} \quad (8)$$

$$0 \leq (p_{r,t}^{\text{PV}})^2 + (q_{r,t}^{\text{PV}})^2 \leq (s_r^{\text{PV}})^2, r \in \mathcal{R} \quad (9)$$

D. Battery Constraints

The battery constraints include active power and capacity limitations [31]:

$$p_b^{\text{Battery,min}} \leq p_{b,t}^{\text{Battery}} \leq p_b^{\text{Battery,max}}, b \in \mathcal{B} \quad (10)$$

$$e_{b,t+\Delta t} = e_{b,t} - p_{b,t}^{\text{Battery}} \Delta t, b \in \mathcal{B} \quad (11)$$

$$e_b^{\text{min}} \leq e_{b,t} \leq e_b^{\text{max}}, b \in \mathcal{B} \quad (12)$$

E. Voltage Constraints

Voltage constraints require that the node voltages remain within a safe range:

$$v_{j,t}^{\text{min}} \leq v_{j,t} \leq v_{j,t}^{\text{max}}, j \in \mathcal{C} \quad (13)$$

The above traditional OPF model relies on accurate physical parameters and power flow calculations, which are difficult to obtain in practice [35]. With the advancement of measuring and communication technologies, enormous amounts of historical measurement data and corresponding topology data can be collected [3], [14]. This enables us to use data-driven methods to estimate node voltages, replacing the traditional power flow calculations (2) ~ (6). The general form of data-driven voltage estimation can be expressed as

$$\bar{v}_t = \mathbb{M}(\mathbf{x}, \mathcal{P}) \quad (14)$$

where $\mathbf{x} = [p^{\text{PV}}, q^{\text{PV}}, p^{\text{Battery}}, p^{\text{EV}}, p^{\text{Load}}, q^{\text{Load}}, q^{\text{Other}}]^T$ is the input data of $\mathbb{M}$ from historical measurement data; $\Lambda = [p^{\text{PV}}, q^{\text{PV}}, p^{\text{Battery}}]^T$ is the decision variables of the OPF; $p^{\text{PV,MPPT}}$, p^{EV} , p^{Load} and q^{Load} are measured values and are treated as constants in the OPF; q^{Other} is determined in the day-ahead scheduling. Due to uncertainties caused by measurement noise and estimations, it is essential to quantify estimation errors of $\bar{v}_t$ to ensure reliable voltage regulation. It should be noted that the method proposed in this paper can easily integrate other devices with continuous control ranges, similar to PVs. However, it is not compatible with devices requiring on/off or stepwise control, which would need to be solved using mixed-integer programming.

III. PROPOSED CHANCE-CONSTRAINED VOLTAGE REGULATION METHOD

This section first shows the background of the NNGP method and then derives the NNGP-based voltage chance constraint. In the proposed method, NNGP serves as a surrogate model for model-based power flow calculations used for node voltage estimation and is fully integrated into the non-linear non-convex OPF framework.

A. NNGP-Based Probabilistic Node Voltage Estimation

DNNs and GPs have shown great advantages in data fitting and modeling in recent years. The equivalence between the two has been extensively studied. [36] proves that a single-layer fully-connected neural network with an i.i.d. prior over its parameters is equivalent to a GP, in the limit of infinite network width. This correspondence enables exact Bayesian inference for infinite width neural networks on regression tasks by means of evaluating the corresponding GP. [36] further suggested that a similar relationship could potentially apply to deeper networks. In addition, kernel functions which mimic multi-layer random neural networks have been developed. Expanding upon previous research, NNGP establishes the precise equivalence between infinitely wide deep networks and GPs, creates a computationally efficient method to compute the covariance function for these GPs, and performs Bayesian inference for wide DNNs [13]. In this paper, NNGP is used to estimate the node voltage in ADSs.

For a L -hidden-layer fully-connected DNNs with each hidden layer l having a width of N_l , the l -th layer can be expressed as

$$z_i^l(x) = b_i^l + \sum_{j=1}^{N_l} W_{ij}^l x_j^l(x) \quad (15)$$

$$x_j^l(x) = \phi(z_j^{l-1}(x)) \quad (16)$$

where W_{ij}^l and b_i^l are independent and randomly drawn and set to zero mean and variance σ_w^2/N_l and σ_b^2 , respectively. As $z_i^l(x)$ is the sum of i.i.d. random terms, according to multidimensional Central Limit Theorem [37], when $N_l \rightarrow \infty$, any finite collection $\{z_i^1(x^{\alpha=1}), \dots, z_i^1(x^{\alpha=k})\}$ (α denotes different samples) will have joint multivariate Gaussian distribution, i.e., $z_i^l \sim \mathcal{GP}(0, K^l)$. The covariance K^l is

$$K^l(x, x') \equiv \mathbb{E}[z_i^l(x) z_i^l(x')] = \sigma_b^2 + \sigma_w^2 \mathbb{E}_{z_i^{l-1} \sim \mathcal{GP}(0, K^{l-1})} [\phi(z_i^{l-1}(x)) \phi(z_i^{l-1}(x'))] \quad (17)$$

where x and x' are two different dataset. By induction, the expectation in (17) is equivalent to integrating against the joint distribution of $z_i^{l-1}(x)$ and $z_i^{l-1}(x')$, which can be described by a zero mean, two-dimensional Gaussian with covariance matrix elements $K^{l-1}(x, x')$, $K^{l-1}(x, x)$, and $K^{l-1}(x', x')$. Therefore, the shorthand of (17) is

$$K^l(x, x') = \sigma_b^2 + \sigma_w^2 F_\phi(K^{l-1}(x, x'), K^{l-1}(x, x), K^{l-1}(x', x')) \quad (18)$$

where the relationship between K^l and K^{l-1} can be recursively derived using the function F_ϕ , whose form depends only on the nonlinearity ϕ . By a series of iterative computations, K^L for the GP describing the network's final output can be obtained. Initially, we can utilize the recursion relating K^1

and K^0 . The base case K^0 with $W_{ij}^0 \sim \mathcal{N}(0, \sigma_w^2/d_{in})$ and $b_j^0 \sim \mathcal{N}(0, \sigma_b^2)$ is

$$K^0(x, x') = \mathbb{E}[z_j^0(x)z_j^0(x')] = \sigma_b^2 + \sigma_w^2 \left(\frac{x \cdot x'}{d_{in}} \right) \quad (19)$$

For certain activation functions, (18) can be calculated analytically using kernel functions to mimic the behavior [38]. For the analytical form of ReLU activation function, the well-known arc-cosine kernel is used, with its detailed derivation provided in [39] (In [39], $J_1(\theta)$, i.e., $n = 1$, is used to mimic the ReLU activation function). The analytical form of ReLU including weight variance σ_w and bias variance σ_b is

$$K^l(x, x') = \sigma_b^2 + \frac{\sigma_w^2}{2\pi} \sqrt{K^{l-1}(x, x)K^{l-1}(x', x')} \quad (20)$$

$$\left(\sin \theta_{x, x'}^{l-1} + \left(\pi - \theta_{x, x'}^{l-1} \right) \cos \theta_{x, x'}^{l-1} \right)$$

$$\theta_{x, x'}^l = \cos^{-1} \left(\frac{K^l(x, x')}{\sqrt{K^l(x, x)K^l(x', x')}} \right) \quad (21)$$

Suppose $\mathcal{D} = \{(x^1, v^1), \dots, (x^n, v^n)\}$ is the historical dataset, where (x, v) is the input-target pair. The distribution of Bayesian estimation for test data x^* using a distribution over functions $z(x)$ is [40]

$$P(z^*|\mathcal{D}, x^*) = \int dz P(z^*|z, x, x^*) P(z|\mathcal{D}) \quad (22)$$

$$= \frac{1}{P(\mathbf{v})} \int dz P(z^*, z|x^*, x) P(\mathbf{v}|z)$$

where $\mathbf{v} \equiv (v^1, \dots, v^n)^T$ represents the targets on the training set; $P(\mathbf{v}|z)$ denotes the observation noise with mean z and variance σ_ϵ^2 . $z^1, \dots, z^n, z^*$ are $n + 1$ samples from a GP. $z^*, z|x^*, x \sim \mathcal{N}(0, \mathbf{K})$ is a multivariate Gaussian with the covariance matrix $\mathbf{K}$ described as

$$\mathbf{K} = \begin{bmatrix} K_{\mathcal{D}, \mathcal{D}} & K_{x^*, \mathcal{D}}^T \\ K_{x^*, \mathcal{D}} & K_{x^*, x^*} \end{bmatrix} \quad (23)$$

where $K_{\mathcal{D}, \mathcal{D}}$ is an $n \times n$ matrix whose (i, j) -th element is $K(x^i, x^j)$ with $x^i, x^j \in \mathcal{D}$, while the i -th element of $K_{x^*, \mathcal{D}}$ is $K(x^*, x^i)$ with $x^i \in \mathcal{D}$; $K_{x^*, \mathcal{D}}$ and K_{x^*, x^*} are defined similarly. By integrating (22), the mean $\bar{\mu}$ and covariance $\bar{K}$ of $z^*|\mathcal{D}, x^* \sim \mathcal{N}(\bar{\mu}, \bar{K})$ are

$$\bar{\mu} = K_{x^*, \mathcal{D}} (K_{\mathcal{D}, \mathcal{D}} + \sigma_\epsilon^2 \mathbb{I}_n)^{-1} \mathbf{v} \quad (24)$$

$$\bar{K} = K_{x^*, x^*} - K_{x^*, \mathcal{D}} (K_{\mathcal{D}, \mathcal{D}} + \sigma_\epsilon^2 \mathbb{I}_n)^{-1} K_{x^*, \mathcal{D}}^T \quad (25)$$

The predicted distribution $z^*|\mathcal{D}, x^*$ is determined through straightforward matrix computations, yet it corresponds to fully Bayesian training of the DNN.

B. Proposed NNGP-Based Voltage Chance Constraint

By using online measured operating data, NNGP can replace parameter-based power flow calculations and achieve node voltage estimations. To reduce the computational burden, estimations and subsequent regulations are performed only on nodes that have had voltage violations in historical data. Specifically, any node with even one record of a voltage violation is considered to be at risk of voltage violations. According to (14) and (24), the voltage estimation results based on NNGP can be described as follows:

$$\bar{\mu}_{j,t}, \bar{K}_{j,t} = \mathcal{NNGP}(\mathbf{x}, \mathbf{P}), j \in \mathcal{V} \quad (26)$$

where $\bar{\mu}_{j,t}$ will replace $v_{j,t}$ in the objective function (1), modifying the voltage term to $\sum_{j \in \mathcal{V}} (\bar{\mu}_{j,t} - 1)^2$. Considering the uncertainties caused by data measurement and estimation, we redefine (13) as a form of chance constraint, i.e.,

$$\mathbb{P}(v_{j,t}^{\min} < \bar{\mu}_{j,t} < v_{j,t}^{\max}) \geq \varepsilon, j \in \mathcal{C} \quad (27)$$

The chance constraint given in (27) specifies that the probability of the NNGP-based voltage estimations falling within the range of $v_{j,t}^{\min}$ and $v_{j,t}^{\max}$ is greater than ε . Furthermore, (27) can be reformulated as

$$v_{j,t}^{\min} < \bar{\mu}_{j,t} - \gamma \bar{\sigma}_{j,t}, j \in \mathcal{C} \quad (28)$$

$$\bar{\mu}_{j,t} + \gamma \bar{\sigma}_{j,t} < v_{j,t}^{\max}, j \in \mathcal{C} \quad (29)$$

where γ is the coefficient used to set different confidence probabilities ε and can be calculated through the integral of the probability density function of the normal distribution of voltage estimations, i.e.,

$$\frac{1}{\sqrt{2\pi}\bar{\sigma}_{j,t}} \int_{\bar{\mu}_{j,t}-\gamma\bar{\sigma}_{j,t}}^{\bar{\mu}_{j,t}+\gamma\bar{\sigma}_{j,t}} \exp\left(-\frac{(\bar{v}_{j,t}-\bar{\mu}_{j,t})^2}{2\bar{\sigma}_{j,t}^2}\right) d\bar{v}_{j,t} \geq \varepsilon, j \in \mathcal{C} \quad (30)$$

When the confidence probability for voltage is chosen as 80%, 90%, 95%, and 99%, the corresponding γ are 1.282, 1.645, 1.960, and 2.575, respectively. Based on the aforementioned NNGP-based probabilistic voltage estimation, a new chance-constrained voltage regulation method, which replaces traditional power flow calculations, has been proposed. The objective function is given by (1), with constraints specified by (7) ~ (12), (28), and (29). It should be noted that voltage is simultaneously regulated in the objective function (1) and constraints (28) and (29). In the objective function, all nodes that have recorded voltage violations according to historical data are targeted to regulate their voltages closer to 1, aiming to reduce the overall number of nodes experiencing voltage violations. This objective function resembles the reward mechanism used in RL to limit the voltage violations [25], [33]. On the other hand, the strict chance constraints defined in (28) and (29) only apply to critical users or nodes that require a reliable voltage security range. In addition, for a real distribution system, imposing voltage limits of 0.95 to 1.05 on all nodes may not have a feasible solution or may greatly increase the probability of non-convergence. Therefore, the proposed method in this paper allows for flexible voltage ranges to be set for different nodes across various scenarios. In contrast, achieving this process in RL would require retraining the model.

Furthermore, the proposed method does not require any additional modifications and can adapt to various topological changes. This is due to the inherent differences in estimation between the NNGP and other deep learning methods. For conventional neural networks, the process typically involves training the network using a training dataset, saving the trained model, and then making estimations based on this saved model. As a result, such networks are limited to scenarios that match the topologies seen in the training set and are not adaptable to new or unseen topologies. However, NNGP does not require training and can perform online inference directly based on measurement data. Therefore, NNGP can adapt to topology changes, with the only requirement being

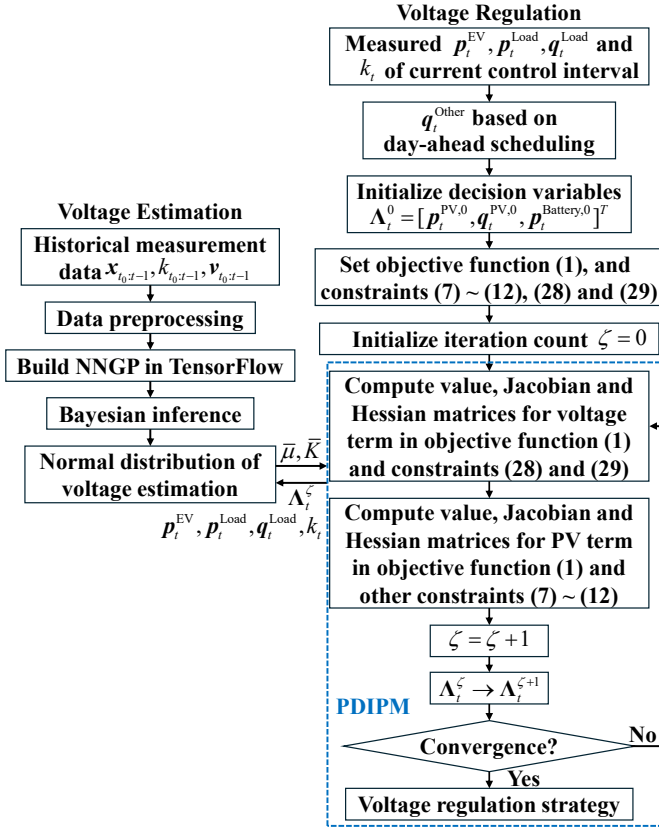


Fig. 1. Proposed analytic NNGP-based chance-constrained voltage regulation framework.

Algorithm 1: NNGP-based probabilistic voltage estimation

Input: Historical measurement dataset $x_{t_0:t-1} = [p_{t_0:t-1}^{PV}, q_{t_0:t-1}^{PV}, p_{t_0:t-1}^{Battery}, p_{t_0:t-1}^{EV}, p_{t_0:t-1}^{Load}, q_{t_0:t-1}^{Load}, q_{t_0:t-1}^{Other}]^T$, $k_{t_0:t-1}$, and $v_{t_0:t-1}$; Decision variables $\Lambda_t = [p_t^{PV}, q_t^{PV}, p_t^{Battery}]^T$; Input data $x_t = [\Lambda_t, p_t^{EV}, p_t^{Load}, q_t^{Load}, q_t^{Other}]^T$; Real-time measured topology k_t

Output: Probabilistic voltage estimation ($\bar{\mu}_t, \bar{K}_t$).

- 1 Data preprocessing for $x_{t_0:t-1}$, x_t and $v_{t_0:t-1}$ according to section III-C.
 - 2 Each $x_{t_0:t-1}$ and $k_{t_0:t-1}$ correspond one-to-one. Based on $k_{t_0:t-1}$, $x_{t_0:t-1}$ is categorized and used for voltage estimation under different topologies.
 - 3 Build NNGP in TensorFlow.
 - 4 NNGP-based probabilistic voltage estimations based on the decision variables Λ_t provided by PDIPM, real-time measured topology k_t and x_t (excluding Λ_t), along with $x_{t_0:t-1}$ and $v_{t_0:t-1}$ corresponding to k_t .
-

the availability of measurement data. In addition, RL-based methods also require retraining when faced with new topologies. In each iteration of the PDIPM, the NNGP estimates voltage through the same online inference process, ensuring adaptability to different topologies.

Algorithm 2: NNGP and PDIPM-based real-time voltage regulation

Input: Real-time measured $p_t^{EV}, p_t^{Load}, q_t^{Load}$ and k_t of current regulation interval; Day-ahead scheduling-based q_t^{Other} .

Output: Optimal voltage regulation strategy, i.e., $\Lambda_t = [p_t^{PV}, q_t^{PV}, p_t^{Battery}]^T$.

- 1 According to the variation range of p^{PV} , q^{PV} , and $p^{Battery}$, set the middle value of that range as the initial iteration value for PDIPM, i.e., $\Lambda_t^0 = [p_t^{PV,0}, q_t^{PV,0}, p_t^{Battery,0}]^T$.
 - 2 Set objective function and constraints according to (1), (7) ~ (12), (28) and (29).
 - 3 Set the initial iteration count of PDIPM to $\zeta = 0$.
 - 4 **while** not reach the maximum iteration number **or** not convergence **do**
 - 5 Construct the input data $x_t^\zeta = [\Lambda_t^\zeta, p_t^{EV}, p_t^{Load}, q_t^{Load}, q_t^{Other}]^T$ for NNGP.
 - 6 According to the details in Algorithm 1, calculate the estimated $\bar{\mu}$ and $\bar{K}$ of nodes voltages. Additionally, use TensorFlow's built-in automatic differentiation functions to calculate the Jacobin and Hessian matrices related to voltage regulation based on (31) ~ (36).
 - 7 Calculate the value, Jacobian and Hessian matrices for the PV term in the objective function (1) and other constraints in (7) ~ (12).
 - 8 $\zeta = \zeta + 1$
 - 9 $\Lambda_t^\zeta \rightarrow \Lambda_t^{\zeta+1}$
 - 10 Verify the convergence criterion [41], [42].
 - 11 **end**
-

C. Algorithm Implementation

The algorithm implementation includes an inner layer of NNGP-based probabilistic voltage estimation and an outer layer of PDIPM-based voltage regulation. The Bayesian inference of NNGP is based on historical data $x_{t_0:t-1} = [p_{t_0:t-1}^{PV}, q_{t_0:t-1}^{PV}, p_{t_0:t-1}^{Battery}, p_{t_0:t-1}^{EV}, p_{t_0:t-1}^{Load}, q_{t_0:t-1}^{Load}, q_{t_0:t-1}^{Other}]^T$, corresponding topology $k_{t_0:t-1}$, and measured voltages $v_{t_0:t-1}$. The input for NNGP includes decision variables $\Lambda_t = [p_t^{PV}, q_t^{PV}, p_t^{Battery}]^T$ from PDIPM. The complete input is $x_t = [\Lambda_t, p_t^{EV}, p_t^{Load}, q_t^{Load}, q_t^{Other}]^T$, along with the topology k_t to determine the historical data $x_{t_0:t-1}$ to be used. Due to the high sparsity of ADSs, data preprocessing is required. The steps include:

1) Based on $x_{t_0:t-1}$, set an intermediate matrix $x_{t_0:t-1}^m$ with the shape $(N_s, 2N_n)$: $x_{t_0:t-1}^m(1:N_s, 1:N_n) = p_{t_0:t-1}^{PV} + p_{t_0:t-1}^{Battery} - p_{t_0:t-1}^{EV} - p_{t_0:t-1}^{Load}$, and $x_{t_0:t-1}^m(1:N_s, N_n+1:2N_n) = q_{t_0:t-1}^{PV} - q_{t_0:t-1}^{Load} + q_{t_0:t-1}^{Other}$. Note: First, the power of each PV, EV, battery, and load needs to be mapped to the corresponding node positions in $x_{t_0:t-1}^m$. Second, the Jacobian and Hessian matrices for PVs, EVs, batteries, and loads at the same position in $x_{t_0:t-1}^m$ are the same.

2) Due to the absence of PVs, batteries, EVs, or loads at many nodes in the distribution grid, numerous columns

in $\mathbf{x}_{t_0:t-1}^m$ consist entirely of zeros. Therefore, to reduce the dimensionality, columns that are entirely zeros are removed, resulting in the final $\mathbf{x}_{t_0:t-1}^f$. It should be noted that $\mathbf{x}_{t_0:t-1}^f$ is derived from $\mathbf{x}_{t_0:t-1}$ through data processing and contains the same data, with the aim of reducing the input dimensionality. $\mathbf{x}_t$ also requires the same data processing procedure. For simplicity, the superscripts m and f will be omitted in subsequent sections of the paper.

3) Similar to $\mathbf{x}_{t_0:t-1}$, reduce the dimensionality of the voltage $\mathbf{v}_{t_0:t-1}$ by eliminating nodes that have not experienced voltage violations in the historical data. For each topology, any node with a record of voltage violations in historical data is considered to be at risk for this topology.

NNGP is fully embedded into the OPF as a surrogate model for voltage estimation, which can be solved using PDIPM. This approach requires the calculation of the Jacobian and Hessian matrices for the objective function and all constraints in each iteration [41], [42]. Jacobian and Hessian matrices are used to determine the search direction and step size for each iteration, which is especially crucial when dealing with nonlinear objective functions and constraints, ensuring that the algorithm iterates effectively towards the optimal solution. The calculation of the Jacobian and Hessian matrices for the PV term in (1), and constraints in (7) ~ (12), is straightforward and can be easily implemented. The Jacobian and Hessian matrices for the voltage term in (1), and the voltage chance constraints in (28) and (29), can be derived as follows (In (28) and (29), all terms are moved to the left-hand side of the inequality symbol $<$)

$$\mathbf{J}_{(1)_V} = 2\gamma_V \sum_{j \in \mathcal{V}} (\bar{\mu}_{j,t} - 1) \frac{d \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}}{d\mathbf{x}^*} \quad (31)$$

$$\mathbf{J}_{(28),j} = -\frac{d\bar{\mu}_{j,t}}{d\mathbf{x}^*} + \gamma \frac{d\bar{\sigma}_{j,t}}{d\mathbf{x}^*}, j \in \mathcal{C} \quad (32)$$

$$\mathbf{J}_{(29),j} = \frac{d\bar{\mu}_{j,t}}{d\mathbf{x}^*} + \gamma \frac{d\bar{\sigma}_{j,t}}{d\mathbf{x}^*}, j \in \mathcal{C} \quad (33)$$

$$\begin{aligned} \mathbf{H}_{(1)_V} = & 2\gamma_V \frac{d \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}}{d\mathbf{x}^*} \left(\frac{d \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}}{d\mathbf{x}^*} \right)^T + \\ & 2\gamma_V \sum_{j \in \mathcal{V}} (\bar{\mu}_{j,t} - 1) \frac{d^2 \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}}{d\mathbf{x}^* d\mathbf{x}^{*T}} \end{aligned} \quad (34)$$

$$\mathbf{H}_{(28),j} = -\frac{d^2 \bar{\mu}_{j,t}}{d\mathbf{x}^* d\mathbf{x}^{*T}} + \gamma \frac{d^2 \bar{\sigma}_{j,t}}{d\mathbf{x}^* d\mathbf{x}^{*T}}, j \in \mathcal{C} \quad (35)$$

$$\mathbf{H}_{(29),j} = \frac{d^2 \bar{\mu}_{j,t}}{d\mathbf{x}^* d\mathbf{x}^{*T}} + \gamma \frac{d^2 \bar{\sigma}_{j,t}}{d\mathbf{x}^* d\mathbf{x}^{*T}}, j \in \mathcal{C} \quad (36)$$

The calculation of the $d\bar{\mu}_{j,t}/d\mathbf{x}^*$, $d\bar{\sigma}_{j,t}/d\mathbf{x}^*$, $d^2 \bar{\mu}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$, $d^2 \bar{\sigma}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$, $d \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}/d\mathbf{x}^*$ and $d^2 \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$ for NNGP can be efficiently implemented using TensorFlow's built-in automatic differentiation functions, and can be further accelerated by GPU. It should be noted that for the objective function, only the calculation of $d \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}/d\mathbf{x}^*$ and $d^2 \sum_{j \in \mathcal{V}} \bar{\mu}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$ needs to be performed. However, for the chance constraint, $d\bar{\mu}_{j,t}/d\mathbf{x}^*$, $d\bar{\sigma}_{j,t}/d\mathbf{x}^*$, $d^2 \bar{\mu}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$, $d^2 \bar{\sigma}_{j,t}/d\mathbf{x}^* d\mathbf{x}^{*T}$ need to be calculated for

each critical node, which will increase the computational burden. However, the calculations for the objective function and each chance constraint are independent, allowing for parallel acceleration using multiple GPUs. The overall computation time will be determined by the most time-consuming calculation among them. It should be noted that in PDIPM, what we actually need are the Jacobian and Hessian matrices of $\bar{\mu}$ and $\bar{\sigma}$ with respect to the decision variables Λ . These can be obtained from the Jacobian and Hessian matrices of $\bar{\mu}$ and $\bar{\sigma}$ with respect to $\mathbf{x}$, since Λ is a subset of $\mathbf{x}$.

In summary, the overall framework is shown in Fig. 1. The detailed steps of NNGP-based probabilistic voltage estimation and voltage regulation are shown in Algorithm 1 and Algorithm 2, respectively. It should be noted that in Algorithm 1, topology k is not a feature that needs to be input into NNGP. Instead, data with the same topology as the real-time measured topology is selected from historical data for Bayesian inference. The classification of samples is solely based on the types of topologies in the historical data, gathering data with the same topology for subsequent Bayesian inference. At the end of this section, we summarize the variables used in this paper:

- Input variables of NNGP (Known + Unknown): $\mathbf{x}_t = [\mathbf{p}_t^{\text{PV}}, \mathbf{q}_t^{\text{PV}}, \mathbf{p}_t^{\text{Battery}}, \mathbf{p}_t^{\text{EV}}, \mathbf{p}_t^{\text{Load}}, \mathbf{q}_t^{\text{Load}}, \mathbf{q}_t^{\text{Other}}]^T$;
- Output variables of NNGP (Unknown): $\bar{\mu}_t$ and $\bar{K}_t$
- Decision variables (Unknown): $\Lambda = [\mathbf{p}_t^{\text{PV}}, \mathbf{q}_t^{\text{PV}}, \mathbf{p}_t^{\text{Battery}}]^T$;
- Measured PV, EV and load power (Known): $\mathbf{p}_t^{\text{PV,MPPT}}, \mathbf{p}_t^{\text{EV}}, \mathbf{p}_t^{\text{Load}}$ and $\mathbf{q}_t^{\text{Load}}$;
- Day-ahead scheduling-based parameters (Known): $\mathbf{q}_t^{\text{Other}}$;
- Topology of current regulation interval (Known): k_t
- Historical measurement data (Known): $\mathbf{x}_{t_0:t-1} = [\mathbf{p}_{t_0:t-1}^{\text{PV}}, \mathbf{q}_{t_0:t-1}^{\text{PV}}, \mathbf{p}_{t_0:t-1}^{\text{Battery}}, \mathbf{p}_{t_0:t-1}^{\text{EV}}, \mathbf{p}_{t_0:t-1}^{\text{Load}}, \mathbf{q}_{t_0:t-1}^{\text{Load}}, \mathbf{q}_{t_0:t-1}^{\text{Other}}]^T$, $k_{t_0:t-1}$ and $\mathbf{v}_{t_0:t-1}$.

IV. TESTING RESULTS

The proposed method is tested on a real unbalanced three-phase 759-node distribution system, which has 120 PVs, 99 batteries, 71 EVs, 163 loads, and 20 switches. The system is located in western Colorado, U.S. Load profile and solar irradiance coefficient are gathered from the yield in 2019, with 15 minutes as the time resolution. The active power output of the PVs varies randomly between 0% and 100%. Similarly, the active power demand of the loads fluctuates randomly between 0% and 100%, with a fixed power factor of 0.98. The EV is either in an on or off state. The active power output or consumption of batteries ranges from the maximum charging power to the maximum discharging power. Seven different feasible topologies are generated by adjusting different switches. Each topology corresponds to 10,000 random operating scenarios. A total of 70,000 samples are generated using OpenDSS to simulate the historical measurement dataset, of which 80% are used as the inference set for Bayesian inference and 20% are used as the test set. It should be noted that for each topology, if a node has even one record of a voltage violation (outside the range of 0.95 to 1.05) within the corresponding 10,000

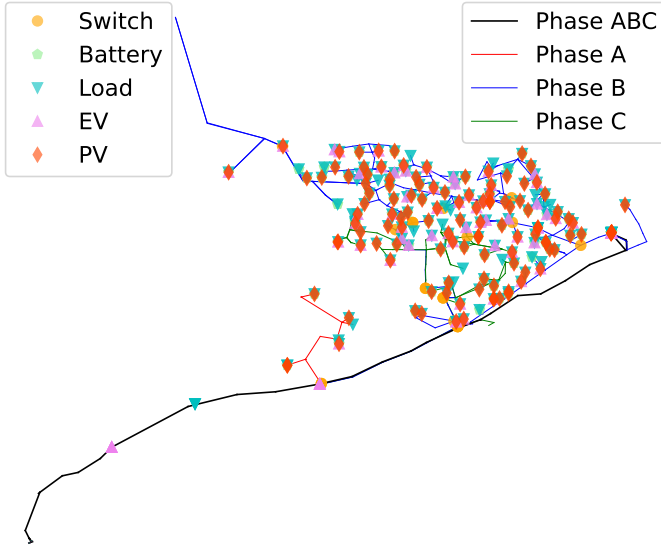


Fig. 2. One-line diagram of the real 759-node distribution system.

samples, it is considered to be at risk of voltage violations and needs to participate in voltage regulation for this topology. The one-line diagram of the 759-node distribution system is shown in Fig. 2.

A. NNGP-Based Probabilistic Voltage Estimation

NNGP is built on the TensorFlow framework [13]. Different from other deep learning methods, NNGP does not require a traditional training process, as most of the hyperparameters are fixed. σ_w and σ_b are set to 1.6, 0.1, respectively. Other hyperparameters are kept the same as in the [13]. In our experiments, we evaluated the performance of NNGP by varying the number of layers from 1 to 10. The MAE, MD, FA and CRPS [43], [44] are calculated as follows

$$\text{MAE} = \frac{1}{N_t N_v} \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{V}} |\bar{v}_{ij} - v_{ij}^*| \quad (37)$$

$$\text{MD} = \frac{1}{N_t N_v} \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{V}} (\bar{v}_{ij} \in \kappa) \text{ and } (v_{ij}^* \notin \kappa) \quad (38)$$

$$\text{FA} = \frac{1}{N_t N_v} \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{V}} (\bar{v}_{ij} \notin \kappa) \text{ and } (v_{ij}^* \in \kappa) \quad (39)$$

$$\text{CRPS} = \frac{1}{N_t N_v} \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{V}} \int_{-\infty}^{\infty} (F(\bar{v}_{ij}) - H(\bar{v}_{ij} - v_{ij}^*))^2 d\bar{v}_{ij} \quad (40)$$

Typically, the importance ranking of these accuracy metrics is $\text{MD} > \text{FA}$ and $\text{CRPS} > \text{MAE}$. This is because MD may cause voltage safety risks, which are serious, while FA increases the number of nodes that need to be regulated. CRPS is used to evaluate the match between the probability distribution of $\bar{v}_{ij}$ and v_{ij}^* , considering the overall shape of the $\bar{v}_{ij}$. This is superior to MAE, which only compares the mean of $\bar{v}_{ij}$ and v_{ij}^* . The testing results with a different number of layers are shown in Table I. From Table I, it can be observed that because NNGP is based on direct Bayesian inference rather than the training of traditional deep learning

TABLE I
COMPARISON OF NNGP ESTIMATION ACCURACY WITH DIFFERENT NUMBER OF LAYERS

Layer	MAE	MD (%)	FA (%)	MD + FA (%)	CRPS
1	0.00230	0.686	0.390	1.076	0.00231
2	0.00226	0.699	0.370	1.069	0.00247
3	0.00223	0.713	0.356	1.069	0.00246
4	0.00222	0.729	0.348	1.077	0.00238
5	0.00220	0.748	0.339	1.086	0.00228
6	0.00220	0.763	0.333	1.096	0.00217
7	0.00220	0.776	0.330	1.106	0.00207
8	0.00219	0.790	0.326	1.115	0.00198
9	0.00219	0.796	0.322	1.118	0.00191
10	0.00219	0.807	0.320	1.127	0.00184

methods, there is a strict correlation between MD, FA and the number of layers. As the number of layers increases, MAE and FA decrease, while MD increases. Conversely, CRPS first increases and then decreases, with a smaller CRPS indicating more accurate estimations. Based on the previous discussion, MD is considered the most important metric because it directly relates to regulation safety. When the number of layers is 1, MD is the smallest, CRPS is relatively low, and the model structure is the simplest, which is advantageous for improving voltage estimation and voltage regulation speed. Therefore, the number of layers is set to 1. It should be noted that choosing a single layer is not because it performs best on all metrics, but simply because we consider MD to be the most important factor. In practical applications, the number of layers can be chosen based on specific circumstances.

Apart from NNGP, we have trained several other data-driven models for comparison, including GCN, GAT [45], DNN, DBN [46] and DGP [47]. It should be noted that DNN, DBN, and DGP are incapable of handling changes in network topology. Therefore, separate models are trained for each topology, and the average accuracy is computed. The comparison results are shown in Table II. From Table II, it can be seen that compared to other data-driven methods, NNGP exhibits clear advantages in terms of MAE, MD, and FA. It is important to note that DNN and DBN cannot handle changes in network topology. Additionally, DNN, DBN, GCN, and GAT are unable to quantify estimation uncertainty. While DGP can quantify estimation uncertainty, it is difficult to train and exhibits low accuracy in multi-output scenarios. By contrast, the advantages of NNGP are evident. The MAE, MD, MD+FA, and CRPS values for NNGP are only 58.5%, 46.2%, 51.9%, and 0.965% of the best values achieved by other data-driven methods, respectively. Additionally, NNGP enables fast and accurate Bayesian inference under different network topologies, with an evaluation time of only 0.47s, meeting the requirements for real-time applications.

Furthermore, for a random operating scenario, the voltage estimation comparisons under two different network topologies using DBN and NNGP are shown in Fig. 3. From Fig. 3, it is evident that the estimated values $\bar{v}$ of NNGP closely align with the true labels v^* for both topology 4 and topology 5, with an MAE of 0.000803 and 0.000571, respectively. Conversely, the estimated values $\bar{v}$ of DBN show significant deviations

TABLE II
COMPARISON OF ESTIMATION ACCURACY BETWEEN NNGP AND OTHER MODELS

Type	Model	Layer	Hidden channel	Head	MAE	MD (%)	FA (%)	MD + FA (%)	CRPS
Traditional method	DNN	4	300	/	0.00422	1.788	0.390	2.178	/
	DBN	4	400	/	0.00393	1.485	0.590	2.075	/
Graph neural network	GCN	6	300	/	0.01283	5.906	1.580	7.486	/
	GAT	7	300	5	0.00848	1.693	0.911	2.604	/
Gaussian process	DGP	4	100	/	0.02362	8.861	6.878	15.739	0.2394
Proposed method	NNGP	1	/	/	0.00230	0.686	0.390	1.076	0.00231

Head is a hyperparameter of GAT, which refers to the number of attention heads.

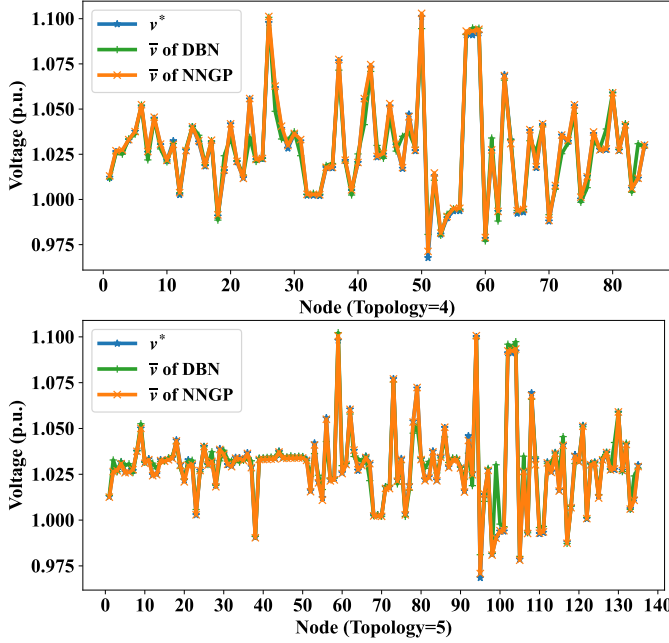


Fig. 3. Comparison of voltage estimation results between DBN and NNGP. **First:** Voltage estimation results of topology 4. **Second:** Voltage estimation results of topology 5.

from the true labels v^* in certain nodes, resulting in an MAE of 0.00399 for topology 4 and 0.00427 for topology 5. It is important to note that in topology 4, there are 85 nodes with potential safety risks based on historical data, while in topology 5, there are 135 such nodes. The purpose is to focus the estimation and voltage regulation efforts on the nodes that truly require regulation, thereby reducing the computational burden.

According to (28) ~ (30), the confidence probability of voltage estimation depends on the selection of γ . By setting γ to 1.282, 1.645, 1.960, and 2.575, the confidence probabilities are 80%, 90%, 95% and 99%, respectively. For the identical random scenario illustrated in Fig. 3 with topology 5, the estimation results under four different confidence levels are shown in Fig. 4. From Fig. 4, it can be observed that as the confidence probability increases, the corresponding confidence intervals become larger, and the probability of v^* falling within the confidence interval also increases. The normal distribution of voltages in Fig. 4 are shown in Fig. 5. Fig. 5 shows that although the mean values of the voltage normal distribution for most nodes fall within the 0.95 to 1.05 range, many exceed

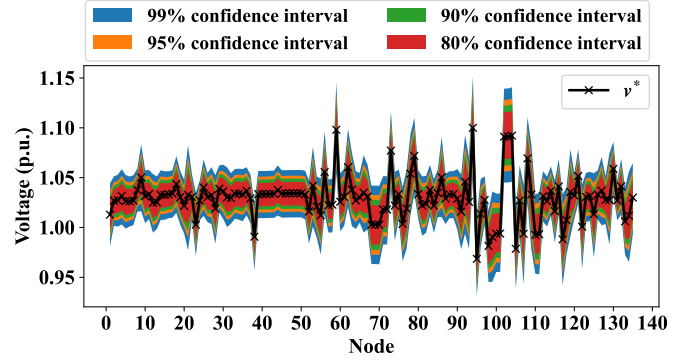


Fig. 4. Probabilistic voltage estimation results.

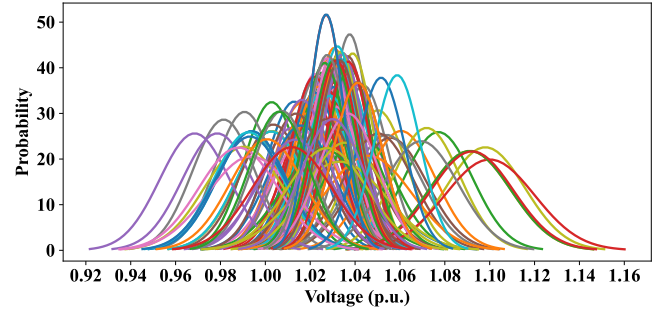


Fig. 5. Normal distribution of voltages in Fig. 4.

1.05, indicating a high risk of over-voltage. Additionally, the voltage normal distribution curves for some nodes far exceed 1.05, making it challenging to find feasible voltage regulation solutions that keep the voltages of all nodes within the 0.95 to 1.05 range. Therefore, in actual power systems, it is necessary to set voltage constraint ranges and critical nodes according to real-world conditions.

B. Voltage Regulation Without Voltage Chance Constraint

In this section, the focus is on optimizing all voltages collectively towards 1.0 in the objective function (1), without considering the voltage chance constraints. To prioritize minimizing PV power curtailment, the parameter γ_{PV} is set to 10, while γ_V is set to 1. Subsequently, continuous voltage regulation over a 24h period is performed using measured profiles of PVs, EVs, and loads at 15 min intervals. The PV profile is consistent for all PVs, as shown in Fig. 8. The EV and load profiles are shown in Fig. 6 and 7, respectively. Fig. 6

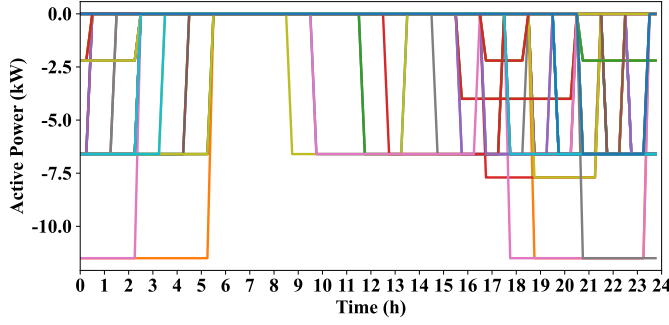


Fig. 6. EV profiles.

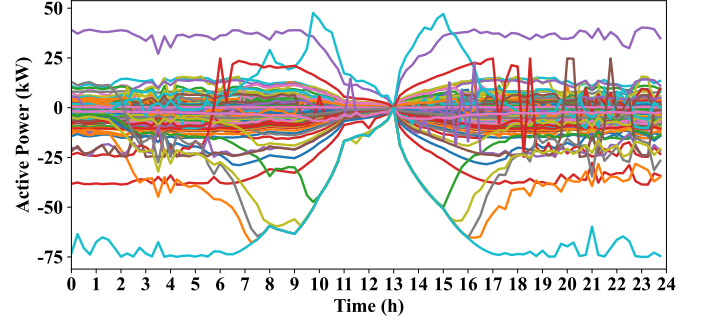


Fig. 9. 24h curve of the PV reactive power.

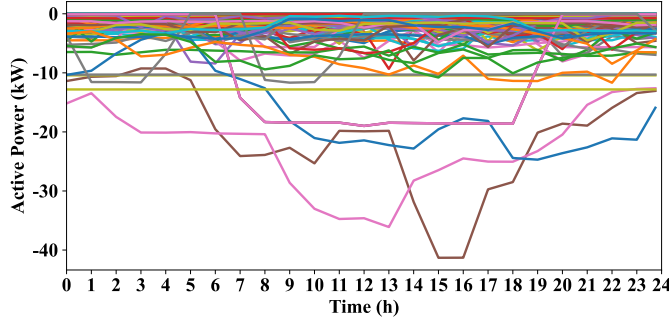


Fig. 7. Load profiles.

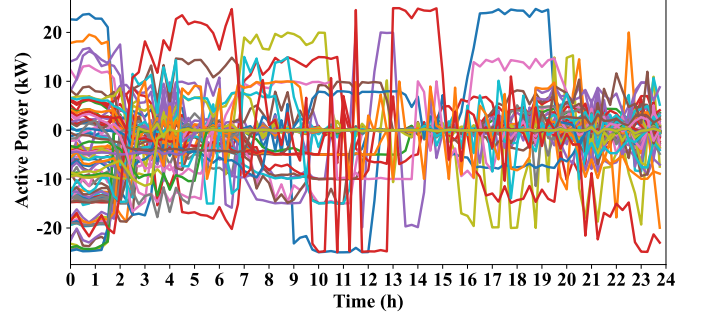


Fig. 10. 24h curve of the battery active power.

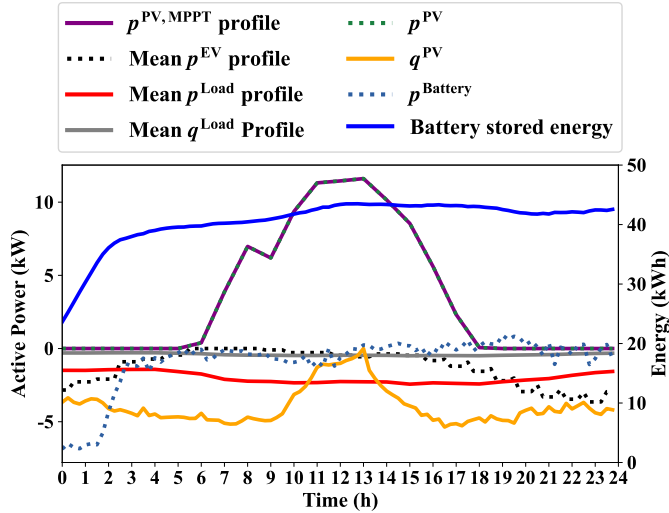


Fig. 8. 24h voltage regulation strategies without voltage chance constraint.

shows that EV charging exhibits a degree of randomness, with sudden increases and decreases in charging power, which can cause voltage fluctuations at local nodes. In Fig. 7, most of the load powers are less than 10 kW and include some constant loads. A few loads have high power and follow a similar daily load profile. Note that in this section, injecting power into the distribution network is considered positive, while absorbing power is considered negative.

The PDIPM-based voltage regulation strategies for one day are illustrated in Fig. 8. From Fig. 8, it can be observed that

the 24h curve of p^{PV} at 15 min intervals follows the $p^{PV,MPPT}$ profile, indicating that the PV term in the objective function (1) is minimized to 0 and there is no PV curtailment during the 24h operation. Specifically, the reactive power curves for each PV and active power curves for each battery corresponding to Fig. 8 are shown in Fig. 9 and 10, respectively. Fig. 9 shows that for most of the time, PV systems either inject or absorb reactive power, improving the voltage levels at surrounding nodes. However, at the 13th hour, q^{PV} decreases to 0. This is because at that time, p^{PV} reaches its maximum value, and the q^{PV} is constrained to 0 according to (9), which limits the voltage regulation capability through q^{PV} at midday. In addition, we can observe that PV has the maximum active power output at midday, while EV has the maximum charging power at nighttime, causing a power temporal and spatial mismatch throughout the day, resulting in high voltage during the day and low voltage at night. To mitigate the imbalance between power generation and load, the battery is charged and discharged throughout the day to compensate for the power temporal and spatial mismatches, as shown in Fig. 10. According to the voltage regulation strategy, the voltage profile for the nodes in $\mathcal{V}$ over a 24h period changes from Fig. 11 to Fig. 12. From Fig. 12, we can see that along the time axis, the voltage is higher at midday, which is consistent with the previous discussion. Moreover, the optimization of the voltage term in (1), as quantified by the MAE, improves from 0.0367 in Fig. 11 to 0.0180 in Fig. 12. In addition, the NVO within the 24h period decreases from 1946 in Fig. 11 to 386 in Fig. 12. This represents only 19.8% of the occurrences before the regulation, indicating the effectiveness of the proposed method

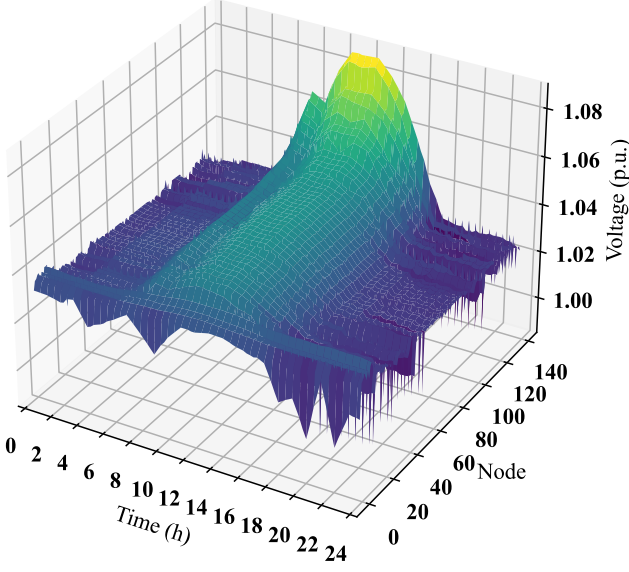


Fig. 11. 24h voltage profiles before regulation (the nodes are sorted by the maximum voltage).

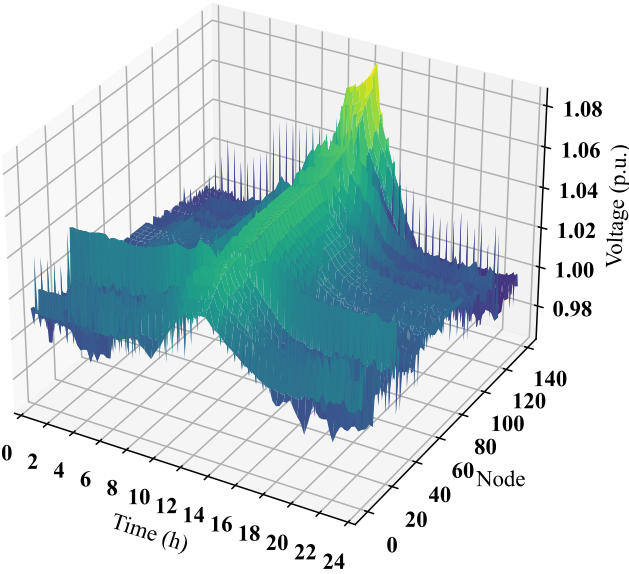


Fig. 12. 24h voltage profiles after regulation without voltage chance constraint (the nodes are sorted by the maximum voltage).

in optimizing the voltage. In each iteration, the total computing time for the voltage term value, Jacobian, and Hessian matrix in the objective function is 5.2s. The total computing time for PDIPM at each time interval is 94s, which satisfies the voltage regulation at 15 min intervals.

C. Voltage Regulation with Voltage Chance Constraint

In this section, it is assumed that a specific node is identified as a critical node, and it is required to strictly regulate the voltage at that node to be within the range of 0.95 to 1.05. Therefore, a voltage chance constraint is added for that node. Using the same operating scenario as described in Section

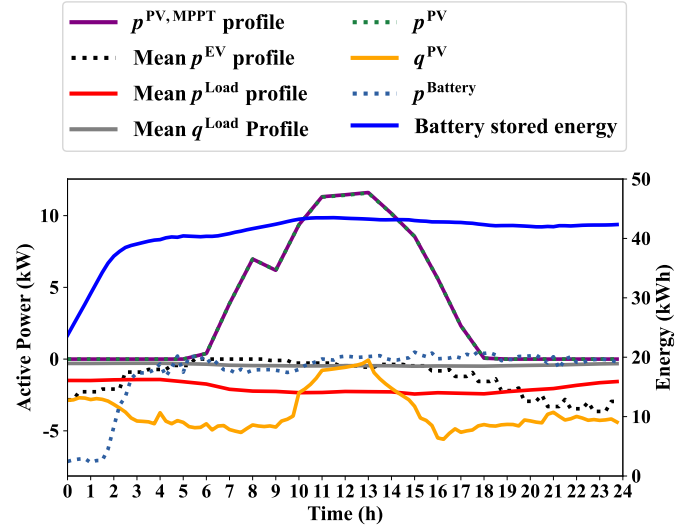


Fig. 13. 24h voltage regulation strategies with voltage chance constraint.

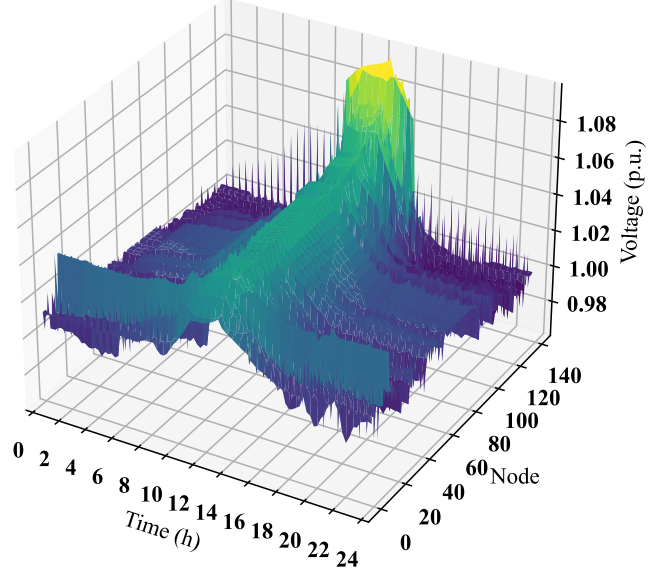


Fig. 14. 24h voltage profiles after regulation with voltage chance constraint (the nodes are sorted by the maximum voltage).

IV-B, a 24h voltage regulation strategy is implemented, and the regulation results are shown in Fig. 13. From Fig. 13, it can be observed that the 24h curve of p^{PV} at 15 min intervals still follows the $p^{PV, MPPT}$ profile, indicating that there is no PV curtailment during the 24h operation. The corresponding voltage profiles after regulation over 24 hours are shown in Fig. 14. Compared to Fig. 12, Fig. 14 shows more voltage peaks during midday. This is because, with the inclusion of chance constraint, the PDIPM prioritizes ensuring the satisfaction of chance constraint at critical nodes, which can result in increased voltage levels at other nodes. The comparison of the voltage profile at the critical node with and without chance constraints is shown in Fig. 15. From Fig. 15, it is evident that after adding the chance constraint, the voltage at the critical node, which was previously exceeding the

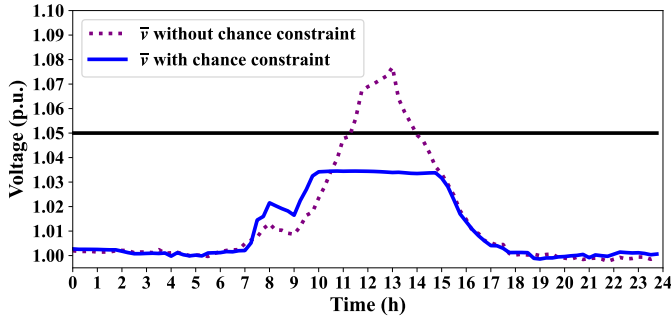


Fig. 15. Comparison of the 24h voltage profile with and without chance constraints.

TABLE III
COMPARISON OF THE PROPOSED METHOD WITH OTHER METHODS

Metric	APL-MPC	IPL-MPC	DBN-VC	IPL-RO	PM
NVO	678	716	544	793	490
MAE	0.0235	0.0234	0.0217	0.0247	0.0186
Time	7.7s	7.2s	275s	86s	384s

limit during midday, remains below 1.05 throughout the 24h period, satisfying the regulation requirements. It is important to note that the $\bar{v}$ ranging from 0.99 to 1.04 provides a safety margin with respect to the limit of 1.05, ensuring a certain level of reliability in the regulation. The NOE and MAE are 490 and 0.0186, respectively. In each iteration, the voltage-related calculations take approximately 24s, resulting in a total computation time of around 384s, which meets the requirement for regulation every 15min.

D. Comparison with Other Voltage Regulation Methods

To show the benefits of the proposed method, five methods are compared: 1) Accurate physical model linearization-based MPC (APL-MPC) [48]; 2) Inaccurate physical model linearization-based MPC (IPL-MPC) [48]; 3) DBN-based voltage regulation (DBN-VC); 4) Inaccurate physical model linearization-based robust optimization considering PV and load uncertainty (IPL-RO) [49]; 5) Proposed method (PM). For the same scenario described in section IV-C, the comparison results are presented in Table III. Additionally, Fig. 16 illustrates the 24h voltage profiles corresponding to the five methods. From Table III, it is evident that the PM demonstrates significant advantages over other methods in terms of NVO and MAE. As compared to methods based on the linearization, the NVO and MAE of PM are only 68.4% and 79.5% of those of IPL-MPC, respectively. Since NNGP provides more accurate estimations than DBN, the NVO and MAE are 90.0% and 85.7% of those for DBN, respectively. Compared to IPL-RO, which relies on linearization or convexification, the NVO and MAE for the PM are 61.8% and 75.3% of those for IPL-RO, respectively. Furthermore, APL-MPC and IPL-MPC, which are based linear power flow, do not require the computation of the Hessian matrix, resulting in significantly faster calculation speeds. However, DBN and the PM have computing times of 275s and 384s, respectively. This difference arises because DBN's training is conducted offline, while NNGP is based

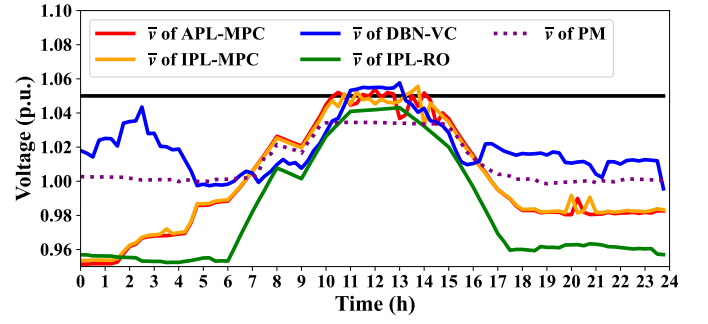


Fig. 16. Comparison of the 24h voltage profile for five methods.

on online Bayesian inference. In addition, it is worth noting that all methods meet the voltage regulation requirement of 15 minutes. From Fig. 16, it can be observed that only the PM and IPL-RO ensure that $\bar{v}$ remains below 1.05 throughout the 24h period. However, because the PM does not assume an accurate nonlinear model, it results in better NVO and MAE compared to IPL-RO, as shown in Table III. In contrast, the other methods exceed 1.05 during midday, posing safety risks.

V. CONCLUSION

This paper proposes an analytic NNGP-based chance-constrained voltage regulation for distribution systems with PVs, batteries, and EVs. The key innovation lies in the utilization of NNGP as a surrogate model for the distribution network physical model, which often struggles to achieve accurate parameters in practice. NNGP can utilize historical data from different distribution network topologies to estimate node voltages and quantify uncertainty, enabling the generation of voltage chance constraints. The derived NNGP can be analytically embedded into the OPF model to perform voltage regulation without the need for an accurate distribution system model assumption compared to existing OPF-based methods. Extensive comparison results obtained on a real 759-node system demonstrate that the proposed method can achieve accurate voltage estimation across different topologies and reliably obtain voltage regulations. The MAE, MD, and MD+FA values for NNGP are only 58.5%, 46.2%, and 51.9% of the best values achieved by other data-driven methods, respectively. Additionally, compared to other voltage regulation methods, the proposed method ensures that the voltage at critical nodes remains below 1.05 throughout the 24h period. In the future, the proposed method will consider the time-coupling charging behavior of EVs, as well as the integration of day-ahead and real-time voltage regulation.

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